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Betsegaw Degefe Agaze

Senior Software Engineer · Backend, Integrations & Cloud

Summary

Backend engineer with 6+ years building integration-heavy systems in regulated domains: national health information exchange (HL7 FHIR, OpenMRS) and fintech (KYC/AML onboarding, crypto custody on AWS). Led end-to-end delivery of Ethiopia's Shared Health Record integration and engineered the Central Data Repository client that aggregates 100K+ patient records from 100+ health facilities. Primary stacks: Node.js/TypeScript (NestJS) and Java (Spring); strong in event-driven and microservice architecture, DDD/CQRS, and CI/CD.

Core Skills

Languages	TypeScript, JavaScript, Java, C#, SQL
Backend	Node.js, NestJS, Next.js (API), Spring Boot, Hibernate, Quarkus, ASP.NET Core, REST, gRPC
Architecture	Microservices, event-driven, DDD, CQRS, multi-tenant systems, OAuth2/Keycloak, mTLS
Cloud & Infra	AWS (Lambda, Step Functions, SQS, DynamoDB, EKS), GCP (BigQuery, Dataform), Docker, Kubernetes, GitHub Actions, Serverless Framework
Data & Messaging	PostgreSQL, MySQL, MSSQL, MongoDB, Redis, Kafka, Elasticsearch, Liquibase
Domain	HL7 FHIR / IPS, HAPI FHIR, OpenMRS 3, OpenFn, OpenCR, KYC/KYB/AML, PEP & sanctions screening, Fireblocks custody, Stripe
Frontend	React, Vue.js, TypeScript, micro-frontends, i18n

Experience

May 2026 – Present
Addis Ababa, Ethiopia

Full Stack Developer
Palladium Global Management LLC – Ethiopia
Full-time · OpenMRS-based national EMR for the Ministry of Health

- Led architecture and delivery of the Shared Health Record (SHR) integration for Ethiopia's national EMR, enabling facilities to exchange HL7 FHIR (IPS) patient summaries with a central HAPI FHIR server.
- Designed the secure exchange pipeline (OAuth2, mTLS, OpenFn workflows) that verifies facility identity, validates national Health IDs, and de-duplicates patients via OpenCR before submission.
- Built multi-tenant ETL and reporting modules (Java, MySQL) that replaced paper MOH clinical registers with automated, exportable reports across multiple facilities.
- Shipped OpenMRS 3 micro-frontends (React, TypeScript) for reporting, ETL, and SHR administration with i18n and unit-test coverage.
- Set up CI/CD from scratch (GitHub Actions, Docker, Maven), enabling same-day releases for backend modules and frontend apps.

Stack Java, Spring, Hibernate, Liquibase, MySQL, React, TypeScript, HL7 FHIR, HAPI FHIR, OpenMRS 3, OpenFn, OpenCR, Keycloak, Docker, GitHub Actions

Dec 2025 – May 2026
Sweden, Remote

Senior Software Engineer (Contract)
Son Of a Coder
Contract · Data platform and CRM product work

- Migrated a legacy Java data pipeline to a three-layer medallion warehouse (Bronze/Silver/Output) on BigQuery with Dataform, replacing hand-written batch jobs with versioned, tested SQL/JS transformations.
- Designed and built an AI-assisted CRM from the ground up: data model, API layer, and LLM-driven lead enrichment and summarisation.

Stack Google Cloud Dataform (SQL + JS), BigQuery, GCP, Node.js, Jest, GitHub

Aug 2025 – Nov 2025
Remote

Independent Consultant
Freelance
Contract · SaaS build for a private client

- Delivered an end-to-end AI meeting-transcription SaaS: NestJS backend, Next.js frontend, PostgreSQL, Stripe subscriptions, and AssemblyAI/Deepgram speech-to-text with speaker diarisation.

Stack NestJS, Next.js, PostgreSQL, Stripe, AssemblyAI, Deepgram

Dec 2024 – Jun 2025
US, Remote

Software Engineer
Yellow Card Financial Inc.
Full-time · Pan-African crypto exchange and stablecoin payments

- Built cloud-native services for real-time digital-asset operations on AWS, covering token transfers, custody, and blockchain monitoring across more than 5 chains.
- Architected serverless APIs and workflows with Lambda, Step Functions, SQS, and DynamoDB, and deployed EKS services that broadcast and monitor on-chain token transfers.
- Integrated Fireblocks for secure custody and signing, and built a Vue.js admin dashboard giving operations real-time visibility into transaction state.

Stack AWS (Lambda, Step Functions, SQS, DynamoDB, EKS), Serverless Framework, Kubernetes, Elasticsearch, Node.js, TypeScript, Vue.js

Jan 2022 – Sep 2024
Varna, Bulgaria, Remote

Full Stack Developer (Contract)
Sepa Cyber Technologies (IT RUN)
Part-time contract · Fintech onboarding and back-office platform

- Engineered onboarding and back-office microservices in Node.js/NestJS for a multi-tenant financial platform, applying DDD and CQRS.
- Integrated Uqudo (KYC), Kompany (KYB, PEP, sanctions, AML), Kraken (crypto rates), and Free Market into the onboarding flow, automating compliance checks that were previously manual.
- Designed inter-service communication with Kafka, Redis MQ, and gRPC; launched the Consumer Management service for tenant and customer administration.
- Contributed to Java Spring Boot and Quarkus services in the same platform; improved throughput of hot paths by 25% through refactoring and query optimisation.

Stack Node.js, NestJS, Spring Boot, Quarkus, PostgreSQL, Redis, Kafka, gRPC, Docker, DDD, CQRS

Dec 2020 – Dec 2024
Addis Ababa, Ethiopia

Software Developer
ICAP at Columbia University
Full-time · Global public-health organisation; Ethiopia HIV/EMR programme

- Engineered the centrally deployed Central Data Repository (CDR) client service that pulls and consolidates data from 100+ health facilities, handling 100K+ patient records for national reporting and analytics.
- Built FHIR-HL7 conversion using the Firely SDK so facility EMR data could be exchanged with national systems in standards-compliant form.
- Designed and tuned MSSQL and MongoDB schemas to sustain high-volume ingestion from concurrent facility uploads.
- Developed and maintained ASP.NET REST services using the Repository Pattern, keeping the codebase maintainable across a multi-year rollout.

Stack ASP.NET, C#, REST, Firely SDK (FHIR/HL7), MSSQL, MongoDB

Nov 2019 – Dec 2020
Addis Ababa, Ethiopia

Software Developer
Orbit Health Inc
Full-time · Ethiopian EMR vendor

- Built APIs and web services in ASP.NET/C# for Orbit Health's EMR modules used in 5+ facilities.
- Refactored the finance and pharmacy modules and optimised their SQL, cutting key report and lookup times by 60%.
- Integrated ORM and reporting tooling into the platform.

Stack ASP.NET, C#, SQL Server, Entity Framework

Education

Aug 2019
Addis Ababa, Ethiopia

B.Sc., Computer Science
Microlink Information Technology College
Graduated with Great Distinction

Languages

Amharic	Native
English	Professional working proficiency